Scaled Partial Pivoting Solution (Problem 9.10)

9.10 Given the equations

$$2x_1 - 6x_2 - x_3 = -38$$
$$-3x_1 - x_2 + 7x_3 = -34$$
$$-8x_1 + x_2 - 2x_3 = -20$$

(a) Solve by Gauss elimination with partial pivoting. Show all steps of the computation.

(b) Substitute your results into the original equations to check your answers.

Initialization

Scale vector (largest absolute entry in each row of the original A, sign ignored):

- $s_1 = \max(2, 6, 1) = 6$
- $s_2 = \max(3, 1, 7) = 7$
- $s_3 = \max(8, 1, 2) = 8$

S = [6, 7, 8] (it never changes)

Index vector: L = [1, 2, 3]

Forward Elimination

Step 1: Eliminate x_1

Select the pivot equation by computing $|a_{li,1}| / s_{li}$ for every row:

- Row 1: 2/6 = 0.3333
- Row 2: 3/7 = 0.4286
- Row 3: 8/8 = **1.0** (largest)

Row 3 is the first pivot equation. Interchange L_1 with L_3: **L = [3, 2, 1]**

Update A and B using row 3 as the pivot row:

- $R_2' = R_2 - (-3/-8)R_3 = R_2 - 0.375R_3$
- $R_1' = R_1 - (2/-8)R_3 = R_1 + 0.25R_3$

Row	x_1	x_2	x_3	b
R_1'	0	−5.75	−1.5	−43
R_2'	0	−1.375	7.75	−26.5
R_3 (pivot)	−8	1	−2	−20

Step 2: Eliminate x_2

Select the second pivot equation from the remaining rows ($L_2 = 2$, $L_3 = 1$):

- Row 2: 1.375/7 = 0.1964
- Row 1: 5.75/6 = **0.9583** (largest)

Row 1 is the second pivot equation. Interchange L_2 with L_3: **L = [3, 1, 2]**

Update using row 1 as the pivot row:

- $R_2'' = R_2' - (-1.375/-5.75)R_1' = R_2' - 0.23913R_1'$
- x_3 coefficient: $7.75 - 0.23913(-1.5) = 8.10870$
- RHS: $-26.5 - 0.23913(-43) = -16.21739$

Row	x_1	x_2	x_3	b
R_1' (2nd pivot)	0	−5.75	−1.5	−43
R_2'' (3rd pivot)	0	0	8.10870	−16.21739
R_3 (1st pivot)	−8	1	−2	−20

Step 3: Eliminate x_3

Only one row is left, so there is nothing to compare. The final index vector is **L = [3, 1, 2]**.

Backward Substitution

Following the order in L:

- $x_3 = -16.21739 / 8.10870 =$ **−2**
- $x_2 = = (-43 - (-1.5)(-2)) / (-5.75) = -46 / -5.75 =$ **8**
- $x_1 = = (-20 - (1)(8) - (-2)(-2)) / (-8) = -32 / -8 =$ **4**

How Good Is the Solution? Residual R = AX − B

Computed with the original A and B, as the lecture notes:

Equation	AX	B	r
1	2(4) − 6(8) − (−2) = −38	−38	0
2	−3(4) − 8 + 7(−2) = −34	−34	0
3	−8(4) + 8 − 2(−2) = −20	−20	0

R = [0, 0, 0], so max $|r_i| = 0$ and the solution is acceptable.

Answer

$x_1 = 4$, $x_2 = 8$, $x_3 = -2$

LU Decomposition (Problem 10.9)

10.9 Solve the following set of equations with LU decomposition:

$$3x_1 - 2x_2 + x_3 = -10$$
$$2x_1 + 6x_2 - 4x_3 = 44$$
$$-x_1 - 2x_2 + 5x_3 = -26$$

Step 1: Decompose A = LU

Eliminate column 1:

- $l_{21} = 2/3$
- $l_{31} = -1/3$

Row 2 becomes: $6 - (2/3)(-2) = 22/3$ and $-4 - (2/3)(1) = -14/3$
Row 3 becomes: $-2 - (-1/3)(-2) = -8/3$ and $5 - (-1/3)(1) = 16/3$

Eliminate column 2:

- $l_{32} = (-8/3)/(22/3) = -4/11$

Row 3: $16/3 - (-4/11)(-14/3) = 16/3 - 56/33 = 40/11$

Result:

L:

1	0	0
0.6667	1	0
−0.3333	−0.3636	1

U:

3	−2	1
0	7.3333	−4.6667
0	0	3.6364

Step 2: Forward Substitution (LZ = C)

- $z_1 = -10$
- $z_2 = 44 - (2/3)(-10) = 152/3 = 50.6667$
- $z_3 = -26 - (-1/3)(-10) - (-4/11)(152/3) = -120/11 = -10.9091$

Step 3: Back Substitution (Ux = Z)

- $x_3 = (-120/11)/(40/11) =$ **−3**
- $x_2 = (152/3 - (-14/3)(-3))/(22/3) = (110/3)/(22/3) =$ **5**
- $x_1 = (-10 - (-2)(5) - (1)(-3))/3 = 3/3 =$ **1**

Check

- $3(1) - 2(5) + (-3) = -10$ ✓
- $2(1) + 6(5) - 4(-3) = 44$ ✓
- $-(1) - 2(5) + 5(-3) = -26$ ✓

Answer

$x_1 = 1$, $x_2 = 5$, $x_3 = -3$

Gauss-Seidel Solution (Problem 11.11)

11.11 Use the Gauss-Seidel method to solve the following system until the percent relative error falls below $\varepsilon_s = 5\%$,

$$10x_1 + 2x_2 - x_3 = 27$$
$$-3x_1 - 6x_2 + 2x_3 = -61.5$$
$$x_1 + x_2 + 5x_3 = -21.5$$

Solving a 3×3 linear system with the Gauss-Seidel method until the percent relative error falls below ε_s = 5%. The System

- $10x_1 + 2x_2 - x_3 = 27$
- $-3x_1 - 6x_2 + 2x_3 = -61.5$
- $x_1 + x_2 + 5x_3 = -21.5$

The matrix is diagonally dominant (|10| > 3, |−6| > 5, |5| > 2), so the method converges.

Update Equations

Solve each equation for its diagonal variable:

- $x_1 = (27 - 2x_2 + x_3) / 10$
- $x_2 = (61.5 - 3x_1 + 2x_3) / 6$
- $x_3 = (-21.5 - x_1 - x_2) / 5$

Start with $x_1 = x_2 = x_3 = 0$.

Iteration 1

- $x_1 = (27 - 0 + 0)/10 =$ **2.7**
- $x_2 = (61.5 - 3(2.7) + 0)/6 = (61.5 - 8.1)/6 =$ **8.9**
- $x_3 = (-21.5 - 2.7 - 8.9)/5 = -33.1/5 =$ **−6.62**

Iteration 2

- $x_1 = (27 - 2(8.9) + (-6.62))/10 = (27 - 17.8 - 6.62)/10 =$ **0.258**
- $x_2 = (61.5 - 3(0.258) + 2(-6.62))/6 = (61.5 - 0.774 - 13.24)/6 =$ **7.9143**
- $x_3 = (-21.5 - 0.258 - 7.9143)/5 =$ **−5.9345**

Percent relative errors:

- $\varepsilon_1 = |(0.258 - 2.7)/0.258| = 946.5\%$
- $\varepsilon_2 = |(7.9143 - 8.9)/7.9143| = 12.45\%$
- $\varepsilon_3 = |(-5.9345 + 6.62)/(-5.9345)| = 11.55\%$

Iteration 3

- $x_1 = (27 - 2(7.9143) + (-5.9345))/10 = (27 - 15.8286 - 5.9345)/10 =$ **0.52369**
- $x_2 = (61.5 - 3(0.52369) + 2(-5.9345))/6 = (61.5 - 1.57107 - 11.869)/6 =$ **8.00999**
- $x_3 = (-21.5 - 0.52369 - 8.00999)/5 =$ **−6.00674**

Percent relative errors:

- $\varepsilon_1 = |(0.52369 - 0.258)/0.52369| = 50.73\%$

- $\varepsilon_2 = |(8.00999 - 7.9143)/8.00999| = 1.19\%$
- $\varepsilon_3 = |(-6.00674 + 5.9345)/(-6.00674)| = 1.20\%$

Iteration 4

- $x_1 = (27 - 2(8.00999) + (-6.00674))/10 = (27 - 16.01998 - 6.00674)/10 =$ **0.49733**
- $x_2 = (61.5 - 3(0.49733) + 2(-6.00674))/6 = (61.5 - 1.49199 - 12.01348)/6 =$ **7.99909**
- $x_3 = (-21.5 - 0.49733 - 7.99909)/5 =$ **−5.99928**

Percent relative errors:

- $\varepsilon_1 = |(0.49733 - 0.52369)/0.49733| = 5.30\%$
- $\varepsilon_2 = |(7.99909 - 8.00999)/7.99909| = 0.136\%$
- $\varepsilon_3 = |(-5.99928 + 6.00674)/(-5.99928)| = 0.124\%$

x_1 is still above 5%, so continue.

Iteration 5

- $x_1 = (27 - 2(7.99909) + (-5.99928))/10 = (27 - 15.99818 - 5.99928)/10 =$ **0.50025**
- $x_2 = (61.5 - 3(0.50025) + 2(-5.99928))/6 = (61.5 - 1.50075 - 11.99856)/6 =$ **8.00011**
- $x_3 = (-21.5 - 0.50025 - 8.00011)/5 =$ **−6.00007**

Percent relative errors:

- $\varepsilon_1 = |(0.50025 - 0.49733)/0.50025| = 0.58\%$
- $\varepsilon_2 = 0.013\%$
- $\varepsilon_3 = 0.012\%$

All errors are now below 5%.

Summary

Iter	x_1	x_2	x_3	max ε (%)
1	2.7	8.9	−6.62	—
2	0.258	7.9143	−5.9345	946.5
3	0.52369	8.00999	−6.00674	50.73
4	0.49733	7.99909	−5.99928	5.30
5	0.50025	8.00011	−6.00007	0.58

Answer After 5 iterations, all percent relative errors fall below 5%:

$x_1 \approx 0.5$, $x_2 \approx 8$, $x_3 \approx -6$ These match the exact solution $x_1 = 0.5$, $x_2 = 8$, $x_3 = -6$.